CHAPTER

7

MAIN MARKET FORMS AND MARKET MECHANISM

Learning Objectives

The students will learn:

- *The various types of market on the basis of competition.*
- *The distinguishing characteristics of all the different types of market.*
- *The short run and long run equilibrium of the perfect competition market.*
- *The establishment of the equilibrium price and quantity through market mechanism at various elasticity of demand and supply.*
- *The effect of price ceiling and price floor on the equilibrium price.*

Course of Action
Maximum Time = 3:15 Hrs
Maximum Questions = 91

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Main Market Forms **Page No. 121**

Topic-2
Short Run and Long Run Equilibrium Under Perfect Competition **Page No. 130**

Topic-3
Market Mechanism **Page No. 134**

Topic-1

Main Market Forms

Concepts covered: 1. Meaning and types of market–Perfect competition, Monopoly, Monopolistic, Oligopoly, Monopsony. 2. Features of different forms of markets.

Revision Notes

- **Meaning of Market :** Market may be defined as an arrangement where buyers and sellers can meet to facilitate the exchange or transaction of goods and services.
- **Types of Market : On the basis of competition :** (1) Perfect competition, (2) Monopoly, (3) Monopolistic competition, (4) Oligopoly and (5) Monopsony.
- **Perfect Competition :** It is defined as the situation in which large number of sellers sell homogeneous product at uniform price in the market.

Scan to know more about this topic

Perfect competition

- **Characteristics or Features of Perfect Competition :** (1) Large number of buyers and sellers, (2) Homogeneous product, (3) Free entry and exit, (4) Perfect knowledge, (5) Perfect mobility of factors of production, (6) Absence of transportation cost, (7) No selling cost, (8) Uniform prices, and (9) Horizontal average and marginal revenue curves.
- **Two important conclusions of Perfect Competition Market :** (1) Firm is price taker not maker, (2) Perfectly elastic demand curve.
- **Imperfect Competition :** Imperfect competition is a competitive market environment where there are many vendors but in comparison to the perfect competitive market scenario, they sell heterogeneous products in competitive market. Monopoly, Monopolistic Competition and Oligopoly are types of Imperfect Competition markets.
- **Monopoly :** Monopoly is that market situation in which a firm has the sole right over production or sale of the product and it has no competitor. Monopoly is the price maker.
- **Features of Monopoly :** (1) Single seller and large number of buyers, (2) Firm and industry are synonymous, (3) No close substitutes, (4) Closed Entry of New firm, (5) Negatively sloped AR and MR curves, (6) Price discrimination possible, (7) Full control over supply of goods, and (8) Abnormal profit in the long run.
- **Reasons for emergence of Monopoly :** (a) Government licensing (b) Patent Rights (c) Cartel (d) Control on raw materials.
- **Monopolistic Competition :** Large number of sellers selling differentiated products.

Main Market Forms and Market Equilibrium

Perfect Competition

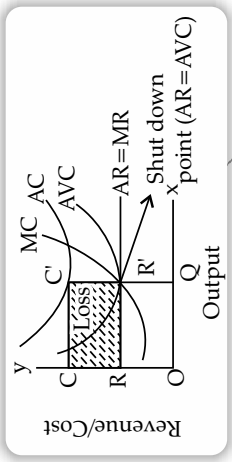
Features

1. Large number of buyers and sellers
2. Homogenous product
3. Free entry or exit of firms
4. Perfect knowledge about product
5. Perfect mobility of factors of production
6. Industry is price maker and firms are price taker
7. Perfectly elastic demand curve
8. Normal profit in long run
9. Absence of selling cost

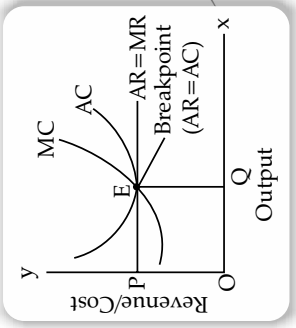
Short Equilibrium of a Firm

Under Short run

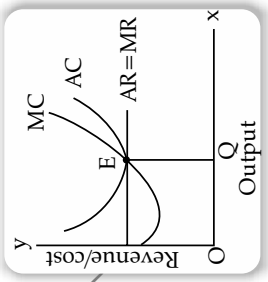
Loss (AR<AC)



Normal Profits (AR=AC)



Under long run (Normal profit AR=AC)



Imperfect Competition

Monopolistic competition

1. Large number of buyers and sellers
2. Product differention
3. Free entry and exit of firms
4. Significant selling cost
5. Lack of perfect knowledge
6. Less mobility of factors of production
7. Non-price competition
8. Downward sloping AR and MR curves (MR < AR)
9. Firms are price maker
10. Normal profit in long run

Oligopoly market

1. Few firms
2. Restricted entry and exit of firm
3. Interdependence of decision-making
4. Non-price competition
5. Price rigidity
6. Indeterminate demand curve
7. Important role of selling cost
8. Formation of cartels
9. Firms are price maker
10. Super normal profits in long run

Monopoly

1. Single seller and large number of buyers
2. No close substitute
3. Barriers to entry and exit of firms
4. Firms have full control over price
5. Price discrimination
6. Downward sloping MR and AR curve (AR > MR)
7. Super normal profits in long run

Trace the Mind Map

- First Level
- Second Level
- Third Level

Market Mechanism

Effect of Change in Supply in case of Demand & Supply

- **Perfectly Elastic Demand**
  - S↑/S↓ →no change in equilibrium price → equilibrium quantity changes
- **Perfectly Inelastic Demand**
  - S↑→P↓→Equilibrium quantity (constant)
  - S↓→P↑→Equilibrium quantity (constant)

Effect of Change in Demand in case of Supply

- **Perfectly Elastic Supply**
  - D↑/D↓→ no change in prices equilibrium quantity changes
- **Perfectly Inelastic Supply**
  - D↑→P↑ equilibrium quantity (constant)
  - D↓→P↓ equilibrium quantity (constant)

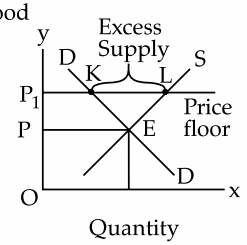
Effect of Simultaneous Change in Demand & Supply

- **Increase**
  - When D↑> S↑ = P↑
  - When D↑ = S↑ = $\overline{P}$
  - When S↑> D↑= P↓
- **Decrease**
  - When D↓> S↓ = P↓
  - When D↓ = S↓ = $\overline{P}$
  - When D↓<S↓= P↑

Applications of Tools of Demand & Supply- (Price Control)

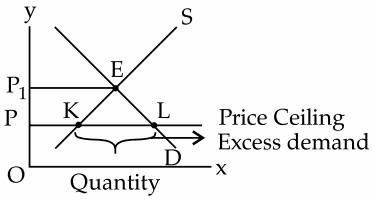
Minimum Price legislation (Floor Price/Minimum Support Price)

- Minimum legal price at which sellers may sell a good
- Government fixed it for agricultural goods
- To predict the interest of farmers
- Set above market equilibrium price
- **Implications**
  - **a.** Situation of excess supply
  - **b.** Surplus output purchased by Government



Maximum Price legislation (Price Ceiling and Rationing)

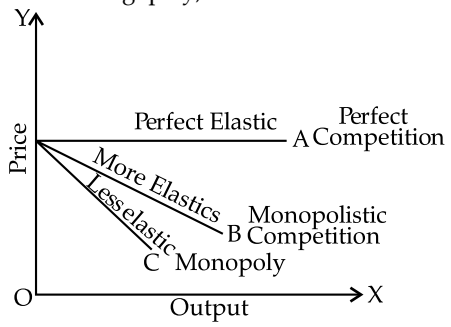
- Maximum legal price to be charged by supplier
- Imposed by Government on essential goods
- To protect the interest of poor consumers
- Set below the equilibrium price
- **Implications**
  - **a.** Situation of Excess Demand
  - **b.** Allocation of available supply
  - **c.** Emergence of Black marketing



Trace the Mind Map

First Level · Second Level · Third Level

- **Characteristics of Monopolistic Competition :** (1) Large number of firms, (2) Product differentiation, (3) Free entry and exit of firms, (4) Selling cost, (5) Non-price competition, (6) Sales techniques, (7) Absence of collective action, (8) Consumer's attachment, (9) Price policy of a firm, (10) Lack of perfect knowledge.
- **Oligopoly :** It is a situation in which there exists few firms producing either homogeneous or differentiated products in a given line of production.
  For example, Coca-cola, Pepsi, etc. soft drink firms come under oligopoly,



- **Main Features of Oligopoly:** (1) Few Sellers, (2) Monopoly Power, (3) Interdependence, (4) Indeterminate Demand curve, (5) Role of selling costs, (6) Lack of uniformity, (7) Price rigidity, (8) Non-price competition, (9) Barriers to entry of firms, (10) Nature of product.
- **Shapes of Firm's Demand Curve Under Different Markets :**

(i) It is a horizontal straight line under perfect competition. It signifies that elasticity of demand $ed = \infty$.

(ii) It slopes downwards under monopoly. Relatively less elastic. This is because there are no close substitutes of the monopoly product in the market.

(iii) It slopes downwards under monopolistic competition but it is relatively more elastic than under monopoly. This is because there is a large number of close substitutes of a product in monopolistic competition.

(iv) It is indeterminate under oligopoly. This is because of a high degree of interdependence between the firms. Price and output policy of one firm significantly impacts the price and output policy of the rival firms in the market.

- **Monopsony :** Monopsony is that market situation in which there is only a single buyer for the commodity. The sole buyer has no competing buyers for the product that he is purchasing from the firm.
  For instance, the government is the single buyer in the purchase of defence equipments from abroad.
- **Features of Monopsony :** (1) Single buyer and large number of sellers, (2) Homogeneous product, (3) Full control over price of goods, and (8) Immobility of factors of production.

Key Terms

- **Homogeneous Product:** The goods which are exactly the same with respect to the size, quality and other physical attributes.
- **Differentiated Product:** The goods which are differentiated with respect to any of the physical attributes of the commodity.
- **Selling Cost:** The cost incurred by the monopolistic competitive firm to sell a product.
- **Non-price Competition:** The competition between the monopolistic competitive firm without an increase or decrease in price but with the help of incurring selling cost.

SUBJECTIVE TYPE QUESTIONS

1 Mark Questions

Q.1. State any one difference between Perfect competition and Monopolistic competition. **[ISC Specimen Paper, 2023-24]**

Ans. Perfect competition market is a type of market where in there are a large number of buyers and sellers selling homogeneous products while Monopolistic Competition Market is a type of market where in there are a large number of buyers and sellers selling differentiated products.

Q.2. Explain one main feature of Monopoly market. **[ISC 2017]**

Concept Applied	Monopoly

Q.3. An oligopolist has an indeterminate demand curve. What is the reason for this feature of Oligopoly? **[ISC Specimen Paper, 2023-24]**

Ans. An oligopolist has an indeterminate demand curve because there is a high degree of interdependence among the firms. Interdependence means that action of one firm affect the actions of other firms.

Q.4. In the monopoly market, there is restriction on entry and exit of firms. What is the reason for this feature of monopoly? **[OEB]**

Ans. In the monopoly market, there is only one firm, hence entry and exit of firms is not allowed. Because one firm controls the whole share of the market.

Q.5. One of the feature of the perfect competition is homogeneous products. What is the reason for this feature of perfect competition? **[OEB]**

Concept Applied	Perfect Competition

Ans. One of the feature of perfect competition is homogeneous products. Homogeneous products are those products which are same in colour, size, design, quality, etc. The reason is that products are close substitutes of each other and have uniform prices.

Q.6. State by giving reason whether the following statement is true or false: **[OEB]**
"Under monopoly, products are differentiated."

This question is for practice and its solution is available at the end of the chapter

Ans. False

Reason: As in monopoly market, product does not have any close substitute. Hence, they are not differentiated.

Q.7. "Under oligopoly market, there is upward sloping demand curve." [OEB]

Concept Applied	Oligopoly

Ans. False

Reason: Under oligopoly market, demand curve is downward sloping but indeterminate. This is because of rigidity of prices.

Q.8. "In the monopoly market, price of the products are uniform." [OEB]

Ans. False

Reason: In monopoly market, since the firm has monopoly power, hence, it can influence price on its own and can charge different prices and thus follow price discrimination.

Q.9. "Under perfect competition, there is restricted entry and exit of firms but under monopoly there is free entry and exit of firms." [OEB]

Ans. False

Reason: This is because under perfect competition there is free entry and exit of firms but under monopoly there is restricted entry and exit of firms.

Q.10. "In the oligopoly market, there are many firms but in monopoly market there are few firms or sellers in the market." [OEB]

Ans. False

Reason: This is because in monopoly market there is only one seller but in oligopoly market there are few firms or sellers in the market.

2 Marks Questions

Q.1. Explain any two causes of restricted entry in a monopoly market structure. [ISC Specimen Paper, 2023-24]

Concept Applied	Monopoly

Ans. The two causes of the restricted entry in a monopoly market structure are:

(i) No close substitutes: The product produced by a monopolist have no close substitutes. So, the monopoly firm has no fear of competition from new or existing products.

(ii) Price Maker: A monopolist is the only seller and himself determines price of his product. He is a price maker. There is no challenge to his price decisions as there are no competitive firms in the market and there are no close substitutes of his product. Barriers to the entry of new firms further strengthens his position as a price maker.

Q.2. The demand curve of a firm is more elastic under monopolistic competition than under monopoly. Explain. A I [ISC Specimen Paper, 2023-24]

Ans. Under monopoly there are no close substitutes of the goods but under monopolistic competition there are close substitutes of the goods in the market. Therefore, monopoly consumers have no choice other than buying the product whereas in the monopolistic competition, close substitution provide a variety of options for the consumers. It makes the demand under monopolistic competition more elastic than under monopoly.

Q.3. Explain the indeterminate demand curve of an oligopolist. [ISC Specimen Paper, 2023-24]

Concept Applied	Oligopoly

Ans. In an Oligopoly form of market, no single firm can predict its prospective sales with perfection. This is because any given change in the price/output decision by a rival firm would initiate a series of actions, reactions and counter actions by others. Therefore, there is no certain nature and position of demand curve under this form of market for a firm.

Q.4. State any two features of Oligopoly. [ISC 2023]

Ans. Refer to question Q4 of 6 Mark Questions Topic 1.

Q.5. What is meant by selling cost? Name one market where selling cost is applicable. [ISC 2020]

Ans. Selling cost: It is the expenditure incurred by the firm to promote the sale of its product through various sales promotion measures such as advertisement in newspapers, TV commercials etc.

It is applicable to a monopolistically competitive firm or to an oligopoly firm.

[ISC Marking Scheme, 2020]

Examiner's Comment

- Many candidates had no idea about selling cost. They simply wrote that it is the price of selling the commodity without mentioning about sales promotion, advertising cost, door to door campaign, etc. Some candidates expressed selling cost as a factor payment cost. A few candidates were not able to explain it with the help of an example. There were candidates who identified the market form as a Monopoly instead of Monopolistic.

Answering Tip

- The students need to understand that the selling cost is incurred to promote sales through newspapers, TV commercials, etc. This is applicable in monopolistic competitive firm or oligopoly.

Q.6. What is meant by price discrimination in monopoly market? [ISC 2018]

Ans. Price discrimination refers to a situation when a producer sells the same product to different buyers at two or more different prices for reasons not associated with difference in the cost of supplying the product to different consumers.

e.g., hospitals charge lower operation fees from the poor patients and higher fees from the rich patients. **[ISC Marking Scheme, 2018]**

Examiner's Comment

- Several candidates defined monopoly market for explaining price discrimination which was irrelevant. Many candidates, instead of writing same product and different prices, wrote different prices are charged for different products.

Answering Tips

- Learn the concept of price discrimination clearly.
- Stress that price discrimination is not associated with difference in cost of supplying the product to different consumers.

Q.7. What is the reason for an indeterminate demand curve under Oligopoly? [ISC 2015]

Ans. Oligopolist has intense competition, cannot ignore the reaction of rival firms as there is Interdependence. **[ISC Marking Scheme, 2015]**

Examiner's Comment

- Some candidates mentioned 'few sellers' instead of 'intense competition' and 'interdependence' in decision making.

Answering Tip

- The implication of each feature of the market must be emphasised. Students should have a clear idea of the kinked demand curve as a reason for price rigidity in oligopoly.

3 Marks Questions

Q.1 Discuss two features of monopoly. [ISC 2015]

Ans. The features of monopoly are:

(i) Single seller and large number of buyers: Monopoly is said to exist when there is only one seller of the product. A monopolist may be the only person, a few partners or in the form of a joint stock company. In simple words, in Monopoly, the number of buyers are large. No one buyer can influence the price by his individual actions.

(ii) Firm and industry are synonymous: There exists only one seller in the case of monopoly market. In such a case, there is no other firm in the industry making the firm and industry one and the same thing.

(iii) No close substitutes: The product produced by a monopolist has no close substitutes. So, the monopoly firm has no fear of competition from new or existing products.

(iv) Closed Entry of New firm: In a monopoly type of market, there is strict barrier on the entry of new firms and no other firm is allowed to enter into the market. Thus, monopolists faces no competition.

(v) Price Maker: A monopolist is the only seller and himself determines price of his product. He is a price maker. There is no challenge to his price decisions as there are no competitive firms in the market and there are no close substitutes of his product. Barriers to the entry of new firms further strengthens his position as a price maker.

(vi) Negatively sloped AR and MR curves: The market demand curve exhibits the total quantity of a particular product that buyers are willing to buy at a specific price. This also helps the monopolist assess the quantity that he can sell at every price that he sets. Therefore, if we assume that a monopolist sets a single price and supplies to all the buyers who want to purchase at that price, then we can find his average and marginal revenues with relative ease.

(vii) Price discrimination possible: In the case of monopoly market there exists a large number of buyers and only a single seller, so the buyers are at the mercy of the seller, letting the seller to enjoy price discrimination. The monopolist can charge different amounts from different buyers.

(viii) Full control over supply of goods: The monopolist can easily influence the supply of the good. During the situation of deficient demand the monopolist can reduce the supply leading the prices to rise abnormally and making the monopolist earn super normal profit.

(ix) Abnormal profit in the long run: In the long run the monopolist always earns supernormal profits.

[Any Two]

Examiner's Comment

- The features of Monopoly were explained correctly by many candidates but in some cases, the quality of the answers was very poor. Some of the candidates misunderstood and wrote about monopolistic competition.

Answering Tip

- Learn the features of various markets with reasons. Highlight key words in the answer.

Q.2 Highlight any three differences between monopolistic competition and oligopoly. [ISC 2014]

Ans.

Monopolistic Competition	Oligopoly
Many sellers	Few sellers
Free entry and exit	Restricted entry
Monopoly in brand and competition from others (as close substitutes).	Intense competition leads to price cutting.

[ISC Marking Scheme 2014]

Examiner's Comment

- Several candidates were confused between 'monopoly' and 'monopolistic competition'. Some wrote the features of these two markets separately, without the points corresponding with each other. They wrote 'closed entry' instead of 'restricted entry' in Oligopoly market against 'free entry and exit' for monopolistic competition, which was incorrect.

Answering Tip

- Students must be clear about the concepts of monopoly and monopolistic competition.

Q.3. In which market, is it not required by the firm to reduce the price to sell more. Draw the TR and MR curve of such a firm. [OEB]

Q.4. Explain the implications of the following features of a Perfectly Competitive Market: [OEB]

(i) Perfect knowledge about the market of the goods

(ii) Perfect knowledge about the inputs used in the production of the goods.

Ans. (i) In a perfectly competitive market, the number of buyers and sellers is very large and all the buyers and sellers have perfect knowledge about the market. As a result no individual buyer or seller can influence the price in the market.

(ii) Another important characteristic of perfect competition is that products are homogeneous and carry the same price. By implication, this means the cost of the inputs used by the producers will be same. As a result of this all sellers have perfect knowledge about the inputs used in the production.

Q.5. Explain the feature 'few firms' and its implications in an oligopoly market. [OEB]

Ans. Oligopoly is a market situation in which there are only a few sellers of a commodity. Under this, each seller can influence its price-output policy. It is because the number of sellers is not very large and each seller controls a big portion of total supply. Price-output policy of a firm does affect the rivals. The price which is fixed under oligopoly without product differentiation is indeterminate. In case of differentiated products, monopoly agreements are even less possible.

Q.6. What difference does it make to the market when we say that there are 'differentiated products' in monopolistic competition? Explain. [OEB]

Concept Applied Monopolistic Competition

Ans. Each firm is in a position to exercise some degree of monopoly, in spite of large number of sellers, through product differentiation. Product differentiation refers to differentiating the products on the basis of brand, size, colour, shape, design, features, etc. The product of a firm is close, but not perfect substitute of other firm.

Implication of 'Product differentiation' is that buyers of a product differentiate between the same products produced by different firms. Therefore, they are also willing to pay different prices for the same product produced by different firms. This gives some monopoly power to an individual firm to influence market price of its product.

4 Marks Questions

Q.1. Explain the implications of the following in an oligopoly market:

(i) Interdependence between firms

(ii) Non-price competition.

Ans. (i) Interdependence between firms :

(a) The most important characteristic feature of oligopoly is interdependence among its firms. The number of sellers is small in this market and each of these firms contribute a significant proportion in the total sales.

(b) As a result, when any one of them undertakes any measure to promote its sale it directly affects other firms and they also immediately react.

(c) Hence, every firm decides its policy after taking into consideration the possible reactions of its rival firms. Thus, every firm is affected by the activities of other firms and it affects others also.

(ii) Non-price competition: When there are only a few firms they are normally afraid of competing with each other by lowering the price. It may start a price war and firm who starts the price war may ultimately lose. Thus, firms indulge in non-price competition in order to increase their share in sales and profit. Under non-price competition they mainly utilise publicity and selling techniques for increasing the sales of their product. Firms under oligopoly follow the policy of rigidity. Price rigidity refers to a situation in which price tends to stay fixed irrespective of changes in demand and supply conditions.

Q.2. Differentiate between oligopoly market and monopolistic competition market. [OEB]

This question is for practice and its solution is available at the end of the chapter

Q.3. State the features of monopolistic competition. [OEB]

Ans. (i) Large number of firms.

(ii) Free entry and exit of firms

(iii) Differentiated products

(iv) AR and MR curves are downward sloping.

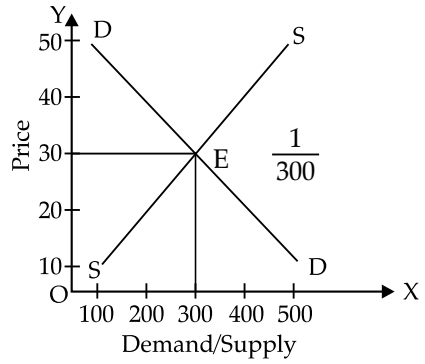
One of the feature of perfect competition is that there are homogeneous products in the market whereas in monopolistic competition there are differentiated products. In perfect competition, products are same in quality, colour, size, design etc whereas in monopolistic competition, products are different in quality, colour, size, design, etc.

Q.4. Explain with the help of a diagram how the price of a commodity is determined in a perfectly competitive market? [OEB]

Ans. In the perfectly competitive market, the price of a commodity is determined by the equality between demand and supply in the industry and hence each and every firm has to take that price as given. Also, in a market where demand curve is downward sloping and supply curve is upward sloping.

Schedule

Price	Demand	Supply	
10	500	100	Demand > Supply
20	400	200	Demand > Supply
30	300	300	Equilibrium
40	200	400	Demand<Supply
50	100	500	Demand< Supply



6 Marks Questions

Q.1. Discuss any four differences between Monopoly and Monopolistic Competition. [ISC 2023]

Ans. The difference between Monopoly and Monopolistic Competition are:

S. No.	Basis of Distinction	Monopoly Market	Monopolistic Competition Market
(i)	Number of Sellers.	There is only one single seller in the market.	There are many sellers in the market.
(ii)	Product Differentiation.	The products sold is completely different from all the products sold in the market.	There is some degree of product differentiation in the market.
(iii)	Selling Cost	No or negligible selling cost has to be incurred by the firm.	A very high selling cost has to be incurred in the market by the firms.
(iv)	Profits in the long run	The firm earns supernormal profit in the long-run	The firms earn normal profit in the long-run
(v)	Freedom of Entry and Exit.	The new firms are restricted to enter into the market.	The new firms are free to enter the market and the old ones are free to exit the market.
(vi)	Demand Curve.	Demand Curve is more inelastic than that of monopolistic competitive firm.	Demand curve is more elastic than that of the monopoly firm.
(vii)	Price Taker/Maker	The firm is a price maker.	The firm is a price taker.

(Any Four)

Q.2. Differentiate between perfectly competitive market and monopolistically competitive market on the basis of the following: [ISC 2020]

(i) Nature of the product (ii) Price influence (iii) Relationship of AR and MR (iv) Demand curve

OR

Discuss four differences between perfect competition and monopolistic competition. [ISC 2016]

Ans.

	Basis	Perfectly competitive market	Monopolistically competitive market
(i)	Nature of the product	Homogeneous	Differentiated

(ii)	Price influence	Price taker. As large number of sellers in the market, a single seller cannot influence the price.	Price maker. Many sellers, but as goods are differentiated some amount of control over the price.
(iii)	Relationship of AR and MR	AR = MR (as price remains same).	AR > MR (as seller reduces price in order to sell more units in the market.)
(iv)	Demand curve	Perfectly elastic demand. Demand curve is parallel to x axis. (large no. of buyers and sellers selling homogeneous goods)	Relatively inelastic/elastic demand. Demand curve is downward sloping. (many sellers selling differentiated goods)

[ISC Marking Scheme, 2020]

Examiner's Comments

- Majority of the candidates did not explain the differences on the basis of the given parameters. Most candidates wrote the differences between the market forms in one or two words.
- Many candidates did not write the implications of each point. Key words were missing in a large number of answer scripts. Candidates wrote, in perfect competition market a seller cannot influence price and in monopolistic competition market he can influence price but the reason for this was not explained by them. Very few candidates could differentiate on the basis of the demand curve.

Answering Tip

- Understand the nature of the product with reference to physical and environmental factors. In case of product differentiation under monopolistic competition relevant examples could be cited. While learning about price influence, stress on the key words. AR and MR curves under both the market forms should be analysed with the help of diagrams. The behaviour of the demand curve should be practiced through diagrams.

Q.3. Explain any four features of perfect competition. [ISC 2018]

Concept Applied | Perfect Competition

Q.4. Explain any four features of oligopoly market. [ISC 2017]

Ans. The features of oligopoly market are:

(i) Few Sellers: Oligopoly is a market situation in which there are only a few sellers of a commodity. Under this, each seller can influence its price-output policy. It is because the number of sellers is not very large and each seller controls a big portion of total supply. Price-output policy of a firm does affect the rivals. The price which is fixed under oligopoly without product differentiation is indeterminate. In case of differentiated products, monopoly agreements are even less possible.

(ii) Monopoly Power: The number of firms is manageable enough to make a guess of the likely reactions of rivals by a firm. There are few sellers of the commodity and each seller has a substantial position in the market. Each firm is aware that it possesses a large degree of monopoly power.

(iii) Interdependence: Interdependence of firms in an oligopoly market—In an oligopoly market, there are small number of big firms. Accordingly, there is a high degree of mutual interdependence. Implying that, price and output policy of one firm has a significant impact on the price and output policy of the rival firms in the market. When one firm lowers its price the rival firms may also lower the price. And, when one firm raises its price, the rival firms may not do so. It is because of the interdependence that it becomes very difficult to estimate change in firm's sales caused by a change in price. Implying that a precise relationship between price and sales cannot be established or, that the firm's demand curve cannot be drawn.

(iv) Indeterminate Demand curve: In an Oligopoly form of market no single firm can predict its prospective sales with perfection. This is because any given change in the price/output decision by a rival firm would initiate a series of actions, reactions and counter actions by others. Therefore, there is no certain nature and position of demand curve under this form of market for a firm.

(v) Role of selling costs: every firm tries to promote its products by different types of expenditures. Advertisement is the most important constituent of the selling cost which affects demand as well as cost of the product. The main purpose of the monopolist is to earn maximum profits; therefore, he adjusts this type of expenditure accordingly.

(vi) Lack of uniformity: The products sold in an oligopoly market are not uniform with respect to the shape, size, cost and price.

(vii) Price rigidity: Price rigidity is the tendency of oligopolist firms to stick to the ongoing price of the product with a view to avoid any sort of price war.

(viii) Non-price competition: When there are only a few firms they are normally afraid of competing with each other by lowering the price. It may start a price war and firm who starts the price war may ultimately lose. Thus, firms indulge in non-price competition in order to increase their share in sales and profit. Under non-price competition they mainly utilise publicity and selling techniques for increasing the sales of their product. Firms under oligopoly follow the policy of rigidity. Price rigidity refers to a situation in which price tends to stay fixed irrespective of changes in demand and supply conditions.

(ix) Barriers to entry of firms: There are various barriers to the entry of new firms. These barriers are almost similar to those under monopoly. Patent rights is the most important form of entry-barriers. Entry of the new firms is extremely difficult, if not impossible.

[Any Four]

This question is for practice and its solution is available at the end of the chapter

Examiner's Comment

- In most of the answers, the two important features of Oligopoly, i.e., 'intense competition' and 'interdependence' were missing.

Answering Tip

- Learn distinguishing features of each market form.

Topic-2 Short Run and Long Run Equilibrium Under Perfect Competition

Concepts covered: Short run equilibrium under perfect competition in three forms–a. losses, b. profits, c. normal profits. Long run equilibrium under perfect competition. (Situation of normal profits). Concept of Break even point, Concept of shut down point.

Revision Notes

- **Price and Output Determination :** The equilibrium of the industry is determined at the point where independent market forces of demand and supply intersect. This equilibrium price is taken by the individual firm. Industry is the price maker and firm is the price taker.

Scan to know more about this topic

Monopoly vs. Oligopoly vs. Competition

- **Short run equilibrium of firm :** Conditions of equilibrium are: (1) AR = MR = MC, (2) MC cuts MR from below.
- **Super normal profits of firm :** A firm may earn short run super normal profits, that is, AR > AC at the point of equilibrium.
- **Super normal losses of firm :** A firm may incur short run super normal losses, that is, AR < AC at the point of equilibrium.
- **Break-even point :** It is the point at which AC = AR. This means, it is the point at which the firm is neither making any profits nor incurring any losses.
- **Shut down point :** It is the point at which the firm is not even able to recover per unit variable cost on sale of a single unit of the commodity or, AVC > AR.
- **Normal profits of the firm :** A firm under perfect competition earns only normal profits in the long run.

SUBJECTIVE TYPE QUESTIONS

1 Mark Questions

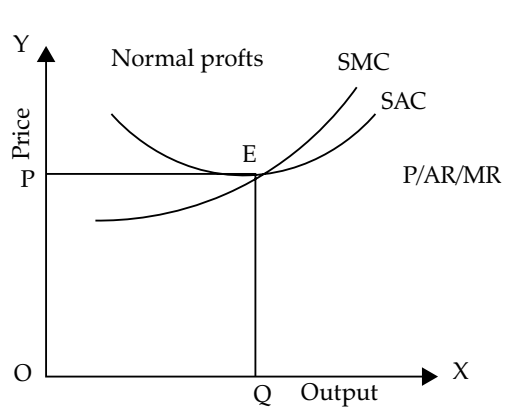
Q.1. Where will the Average Cost curve be in the case of Break – even point? [OEB]

Ans. The Average Cost curve will be tangent to the Average Revenue Curve passing through the point of equilibrium.

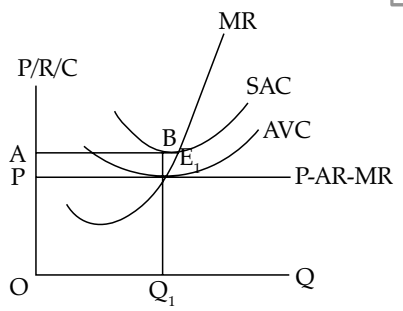
Q.2. Draw the diagram to show break – even point. [OEB]

Concept Applied	Short-Run Equilibrium under perfect competition

Ans.



Q.3. What does the diagram given below show? **[OEB]**



Ans. Shut – down point.

Q.4. Define Shut down point? **[OEB]**

Q.5. Give the meaning of Break-even point? **[OEB]**

Ans. Break-even point can be defined as that point at which average revenue is equal to average cost. Hence, it is that point at which firm is neither earning profits nor incurring losses.

2 Marks Questions

Q.1. What is meant by a break-even point? Show it in a diagram. **[ISC Specimen Paper, 2023-24]**

OR

What is meant by break-even point? Draw a well labelled diagram to show the break-even point. **[ISC 2023]**

OR

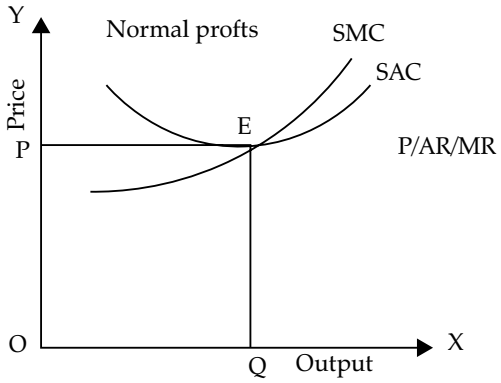
Show with the help of a diagram, how a perfectly competitive firm earns normal profit in short run equilibrium. **[ISC 2019]**

OR

Show with the help of a diagram, how a perfectly competitive firm earns normal profit in short run equilibrium. **[ISC 2015]**

Concept Applied	Short Run equilibrium under perfect competition

Ans. Break – even point or normal profit is the situation when the firm earns that much profit so as to cover the total cost incurred by the firm.



In the figure we can see that the firm attains equilibrium at the point E where the MC = MR and the MC cuts the MR from below. At the equilibrium point the firm gets a total revenue from the area POQE that is by multiplying the Average Revenue of OP by the Output OQ. Likewise we get the total cost incurred by the firm from the area POQE that is by multiplying the Average Cost OP by the Output OQ. The profit earned by the firm is found by finding the difference between the total revenue and the total cost, which in this case is 0. Thus, the firm just break-evens.

Examiner's Comment

- Many candidates did not discuss the derivation of normal profit. Several candidates did not mention the equilibrium conditions. Some candidates drew an incorrect diagram, being confused between AC and MC curves.

Answering Tips

- The students need to practice to draw the diagram stepwise, in order to show how a perfectly competitive firm earns normal profit in short run equilibrium.
- Focus on the fact that the diagram and its explanation are not two separate parts of the answer.
- Learn the definition of normal profit.

Q.2. Under Perfect Competition, a firm will enjoy normal profit in the long run even if it enjoys supernormal profit in the short run. Explain. **[ISC 2023]**

Ans. The firm earns normal profit in the long run even if it enjoys supernormal profit in the short run, under perfect competition. This is due to the freedom of entry and exit in the firm. Say for example, if the firm in the perfect competition earns supernormal profit in the short run, more firms will be attracted to enter the market due to the freedom of entry. This will bring down the prices, so much so that the firms will start earning normal profit.

Q.3. Rita was confused between break - even point and shut down point. Explain to her by clearly differentiating between the two concepts. **[OEB]**

This question is for practice and its solution is available at the end of the chapter

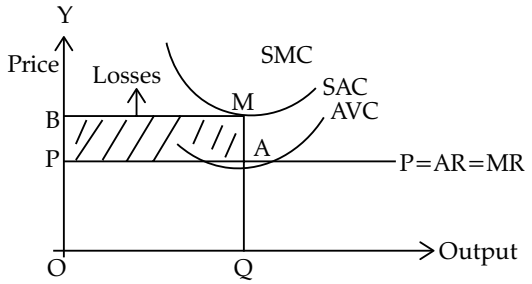
4 Marks Questions

Q.1. Explain the short run equilibrium of a firm facing losses under Perfect Competition. A I **[ISC 2023]**

OR

Explain how a firm in perfect competition incurs loss, in short run equilibrium. [ISC 2018]

Ans. Loss is the situation when the firm is not able to recover the total cost.



We can see in the figure, the firm attains an equilibrium at point A, where the MC = MR and MC cuts the MR from below. At this point the firm earns the total revenue given by the area OPAQ which we get by multiplying the Average Revenue of OP with the output OQ. The Total cost of the firm at the equilibrium point is given by the area OBMQ, which we get by multiplying the average cost OB with the output OQ. As we can see that the firm's total cost is more than the total revenue by the area PBMA, it shows the loss incurred by the firm. But the firm continues to produce as it can still recover the variable cost as the AVC curve is below the price line.

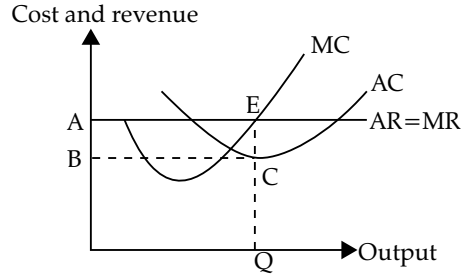
Examiner's Comment

- Several candidates did not draw the AVC curve. Many candidates were unable to plot the correct equilibrium point and the area showing loss. A few candidates drew the price line below the AVC curve.

Answering Tips

- Focus on the fact that the AVC curve has to be drawn in case of situation of loss faced by the firm. The students need to practice to draw a perpendicular from the point where MC cuts MR.
- The students need to understand how to calculate the loss with the help of area of a rectangle.

Q.2. Study the graph given below:



(i) Explain the type of equilibrium that is attained in the graph above.

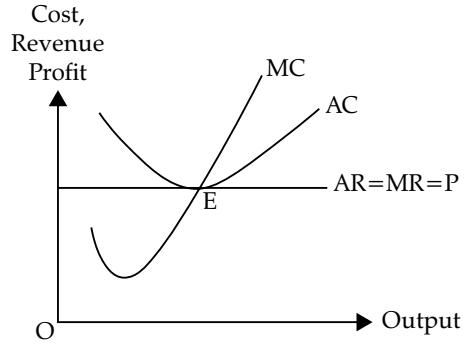
(ii) Explain any one feature of the market shown in the above graph.

Ans. (i) The type of equilibrium shown is super normal profit.

We can see in the figure, the firm attains an equilibrium at point A, where the MC = MR and MC cuts the MR from below. At this point the firm earns the total revenue given by the area OBEQ which we get by multiplying the Average Revenue of OA with the output OQ. The Total cost of the firm at the equilibrium point is given by the area OBCQ, which we get by multiplying the average cost OB with the output OQ. As we can see that the firm's total revenue is more than the total cost by the area BAEC, it shows the super normal profit earned by the firm.

(ii) The market shown in the graph is perfect competition. All the firms produce homogeneous products which are the perfect substitutes of each other.

Q.3. Study the graph given below:



(i) Identify the type of profit earned by the firm.

(ii) Explain with the help of the graph given above, the situation that the firm is facing?

Ans. (i) The firm earns a normal profit.

(ii) Break – even point or normal profit is the situation when the firm earns that much profit so as to cover the total cost incurred by the firm.

6 Marks Questions

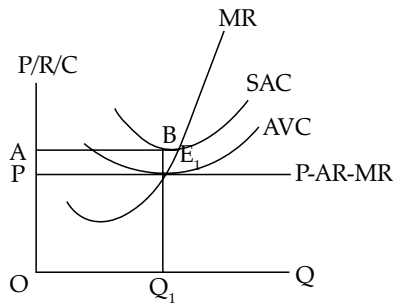
Q.1. Using diagrams, explain the concepts of: A I **[ISC 2020]**

(i) Shut-down point

(ii) Break-even point of a firm in perfect competition under short run.

Ans. (i) Shut-down point.

Shut down point is the situation when the firm is only able to cover its variable cost due to the losses earned by it.



As can be seen in the figure, the firm attains equilibrium at point E_1 where the MC = MR and MC cuts the MR from below. At this point the total revenue earned by the firm is given by the area OPE_1Q_1 that is by multiplying the Average Revenue OP by the Output OQ_1. The total cost incurred by the firm is given by the area $OABQ_1$ that is by multiplying the Average Cost OA by the Output OQ_1. As the total revenue is less than the total cost, the firm suffers loss. Now, when we check the total variable cost, we get it in the area OPE_1Q_1 which is equal to the Total Revenue. Thus, the firm is just able to overcome its variable cost and beyond this it will not be. Thus it will be beneficial for the firm to close its business production.

(ii) Break – even point.

Refer to Question Q1 of 2 mark Question Topic 2.

Examiner's Comments

The common errors made by the candidates while explaining the concept of Shut-down point were:

- Using the TR-TC approach which is incorrect for this concept.
- Not showing the shut-down point at the minimum point of AVC curve.
- Drawing the AR=MR curve as a tangent of AVC curve at its minimum point but not drawing the MC curve and so no equilibrium being established.
- Not defining shut down point. as an equilibrium point which covers only variable cost / where AR=AVC but only mentioning, where firm ceases to operate, which made the answer incomplete.
- Not showing the break-even point at the minimum point of AC curve.
- Having drawn the AR=MR curve as a tangent of AC curve at its minimum point but no MC curve being drawn and so no equilibrium being established.
- Simply defining the break-even point as a situation where either TR=TC or a situation of no profit, no loss without mentioning its position in the diagram. This made the description incomplete.

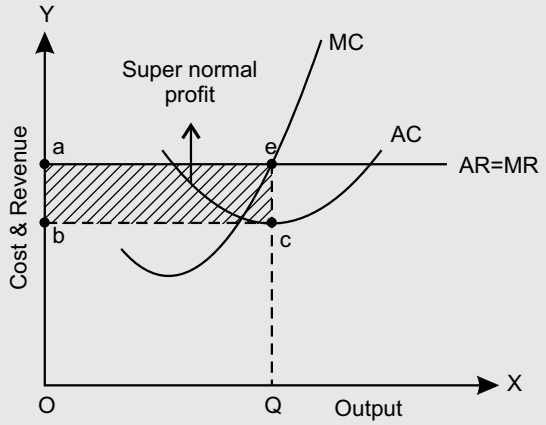
Answering Tip

- Understand through diagrams that Shut-down point and Break-even point are different equilibrium situations of a firm and the diagrams should be completed by establishing equilibrium. Draw separate diagrams for losses and shutdown point and always write the conditions of equilibrium.

Q.2. How does a perfectly competitive firm earn super normal profits in the short run equilibrium? Explain it with the help of a diagram.

[ISC 2016]

Ans. The situation occurs when the price is so high that it covers fully the AC of the commodity and is still above it. The market price is more than AC and the firm earns super normal profits. This situation is graphically illustrated.



At price a determined by the intersection of market demand and supply curve equilibrium is at point e. At point e, MC = MR and MC curve cuts MR from below. Super normal profits are earned.

Super normal profits are calculated as :

AR = MR = P

TR > TC

TR = aeQO

TC = bcQO

Super normal profits = aecb

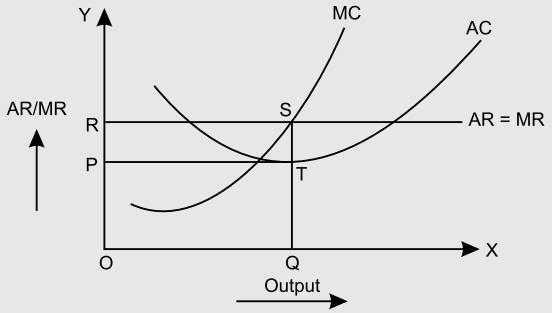
AR = P covers AC

The firm is able to completely cover the AC and still gets price more than AC.

How a competitive firm earns supernormal profit in short run. Short run equilibrium in perfect market.

Super normal profit or excess profit.

Diagram with explanation



Explanation: Equilibrium conditions:

TR = AR X Q = OR X OQ = Area of ORSQ

TC = AC X Q = OP X OQ = Area of OPTQ

TR – TC = Profit = Area of ORSQ – Area of OPTQ = Area of RSTP

Calculation of TR and TC

[ISC Marking Scheme 2016]

Examiner's Comments

- Equilibrium point was not shown correctly by many candidates. Some common errors made by candidates were :
  (i) MC curve did not pass through the maximum point of AC.
  (ii) Perpendicular line from equilibrium to the X axis (output) not drawn.
  (iii) Tangent line to AC on Y axis not shown.

Answering Tip

- Draw line diagrams and explain each step. Equilibrium point and hence equilibrium condition, determination of output from equilibrium point and AC curve should be explained; P>AC, P=AC and P<AC, should be explained to show supernormal profit and loss.

Q.3. "A firm is a price taker and an industry is a price maker under perfect competition." Comment.

[OEB]

Topic-3 Market Mechanism

Concepts covered: Condition of equilibrium. Effect of change in demand on market equilibrium. Effect of change in supply on market equilibrium. Effect of simultaneous change in demand and supply on market equilibrium.

Revision Notes

- **Market Equilibrium :** It is a state in which market demand is equal to market supply.
  For example, the demand of pens is 50 units and the supply of pens is 50 units, so the market is in equilibrium.
- **Equilibrium Price :** It is the price at which market demand is equal to market supply.
- **Equilibrium Quantity :** It is the quantity which corresponds to equilibrium price.
- **Effect of Change in Demand :** Increase in demand raises and decrease in demand lowers the equilibrium price. Also, equilibrium quantity will increase when demand increases and will decrease when demand decreases.
- **In Case of Perfectly Elastic Supply :** Increase or decrease in demand for a commodity does not cause any change in its price in case supply of the commodity is perfectly elastic.
- **In Case of Perfectly Inelastic Supply :** Increase or decrease in demand causes a change in the price of the commodity. Equilibrium quantity remains constant.
- **Effect of Change in Supply :** Increase in supply causes a fall in equilibrium price and decrease in supply causes a rise in equilibrium price. Equilibrium quantity will increase if supply increases and decrease if supply decreases.
- **In Case of Perfectly Elastic Demand :** Increase or decrease in supply does not cause any change in equilibrium price but quantity changes.
- **In Case of Perfectly Inelastic Demand :** Decrease in supply result in an increase in price and increase in supply leads to decrease in price. The equilibrium quantity remains constant.
- **Effect of a Simultaneous Change in Demand and Supply on Equilibrium Price :**
  (i) When demand increase more than supply, equilibrium price will increase.
  (ii) When demand and supply increases equally, equilibrium price remains constant.
  (iii) When supply increases more than demand, equilibrium price falls.
- **Excess Demand:** The situation when the demand for a commodity is more than the supply.
- **Excess Supply:** The situation when the demand for a commodity is less than the supply.

This question is for practice and its solution is available at the end of the chapter

SUBJECTIVE TYPE QUESTIONS

1 Mark Questions

Q.1. When demand and supply increase by equal percentage, what happens to the equilibrium price? [OEB]

Ans. When increase in demand is proportional equal to increase in supply, then rightward shift occurs with demand curve is proportionately equal to rightward shift in supply curve. Consequently, the equilibrium price remains the same. However, the equilibrium quantity rises.

Q.2. Give the difference between the effects of increase in demand and increase in supply on equilibrium price? [OEB]

Ans. Increase in demand can be defined as the situation in which demand increases keeping supply same, hence equilibrium quantity rises and equilibrium price also rises .but in case of increase in supply equilibrium quantity rises but equilibrium price falls but equilibrium quantity rises.

Q.3. State the difference between the effects of decrease in demand and decrease in supply on equilibrium price? [OEB]

Ans. When demand decreases then demand curve shift towards left and hence both equilibrium price and equilibrium quantity falls. But when supply decreases then supply curve shift towards left and hence equilibrium price rises but equilibrium quantity falls.

Q.4. When supply curve is perfectly inelastic, then what will be the effect on equilibrium price and equilibrium quantity if demand increases? [OEB]

Q.5. When demand and supply increase by equal percentage, what happens to the equilibrium price? [ISC Specimen Paper, 2023]

Ans. When increase in demand is proportional equal to increase in supply, then rightward shift occurs with demand curve is proportionately equal to rightward shift in supply curve. Consequently, the equilibrium price remains the same. However, the equilibrium quantity rises.

Q.6. State whether the following statement is true or false with reason: [OEB]
"When there is simultaneous increase in both demand and supply, then equilibrium price remains only constant."

Ans. True
Reason: This is because when there is increase in both demand and supply with equal percentage then equilibrium price remains same.

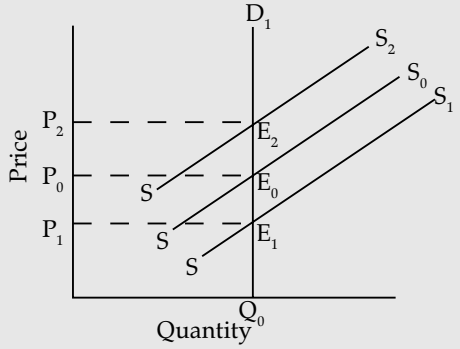
Q.7. "When increase in demand is greater than increase in supply, then equilibrium price falls and equilibrium quantity also falls". [OEB]

Ans. False
Reason: This is because when increase in demand is greater than increase in supply, then equilibrium price rises but equilibrium quantity also rises.

2 Marks Questions

Q.1. Show with the help of a diagram, a situation when change in supply will keep the equilibrium quantity of the commodity unchanged. [ISC 2020]

Ans. Diagram showing change in supply is not affecting equilibrium quantity.



Condition: Perfectly inelastic demand

[ISC Marking Scheme, 2020]

Examiner's Comments

Common errors made by most of the candidates were:

- Drawing a perfectly elastic demand curve.
- Showing a proportional change in demand.
- Showing either an increase or decrease in supply and not both.
- Not taking the perfectly inelastic curve straight up and not labelling it.
- Either showing simultaneous shifting of demand and supply curves, or perfectly inelastic supply curve.
- Drawing only three supply curves and three prices on these supply curves showing the same quantity, but not drawing the vertical demand curve and not mentioning the condition perfectly inelastic demand.

This question is for practice and its solution is available at the end of the chapter

Answering Tip

- Give adequate practice to draw the diagrams so in order to be able to make out the difference between a price line and a line joining the equilibrium points with perfectly elastic demand or supply curves and same for the quantity line with perfectly inelastic demand and supply curves. Well labelled diagrams should be practiced. Also the students need to first draw the perfectly inelastic demand curve and then the supply curves showing equilibrium, quantity remaining unchanged. Supply curves should also show the increase and decrease in supply.

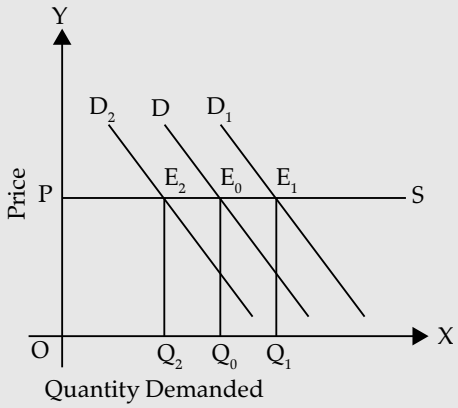
Q.2. With the help of diagrams, show how equilibrium price and quantity of a commodity are affected when: [ISC 2018]

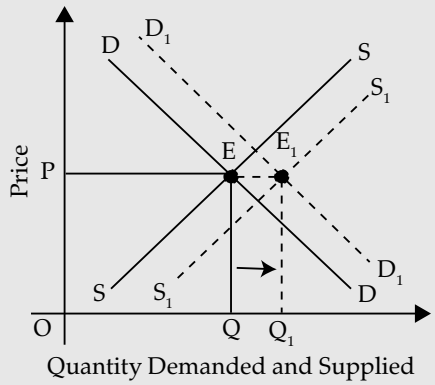
(i) Demand is perfectly elastic and supply decreases.

(ii) Supply is perfectly elastic and demand increases.

Q.3. With the help of a diagram, show how the equilibrium price can remain unchanged even after a rightward shift of the demand curve. [ISC 2016]

Ans.





[ISC Marking Scheme, 2016]

Examiner's Comment

- A common error made by candidates, even by those who drew the correct diagrams was, not labelling the supply curve (horizontal line). Some candidates shifted only the demand curve to the right, without drawing the supply curve.

Answering Tip

- Develop the habit of labelling the diagrams. Equilibrium of market means both demand and supply curves should be present in the diagram.

Q.4. What is maximum price or price ceiling? Explain its implications. [OEB]

Ans. Price Ceiling or Maximum Price : It refers to fixing of the maximum price of a commodity at a level lower than the equilibrium price. Government imposes price ceiling in case of essential commodities (wheat, sugar, kerosene, etc.) when the equilibrium price determined by free market forces of demand and supply is high.

Implications of price ceiling:

(i) Shortage (excess demand): At this controlled price quantity demanded is more than quantity supplied. It creates a shortage and to overcome this shortage government may enforce the rationing system.

(ii) Black marketing: If rationing is not administered by the government effectively it results in black marketing. Due to excess demand buyers will compete and they would be willing to buy a commodity at a higher price than the price fixed by the government.

Q.5. What is minimum support price? [OEB]

Ans. Minimum support price are the prices which is determined by the government for procuring certain crops from farmers to insure them against any sharp fall in prices. It is an integral component of Agriculture Price Policy and it strives to ensure support price to farmers and affordable prices to the consumer. "It is important because all farmers are assured at least the minimum price for their produce".

3 Marks Questions

Q.1. Briefly explain how equilibrium price is determined with the help of demand and supply curves. [ISC 2020]

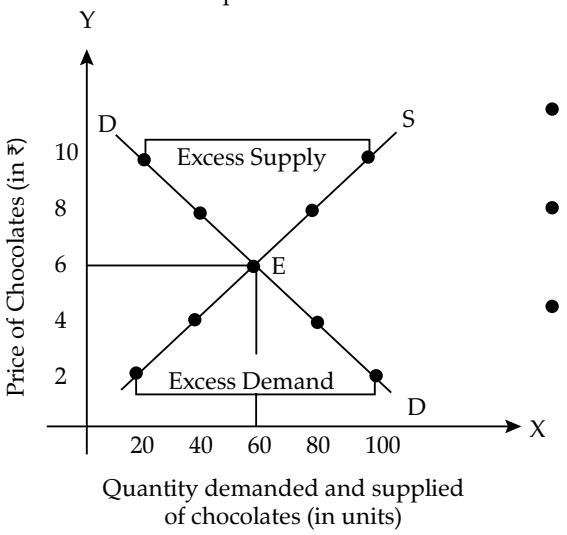
Ans. Market equilibrium is determined when the quantity demanded of a commodity becomes equal to the quantity supplied. The price determined

This question is for practice and its solution is available at the end of the chapter

corresponding to market equilibrium is known as equilibrium price and the corresponding quantity is known as equilibrium quantity.

As we can see in the figure, the equilibrium is attained when the Demand Curve DD intersect the Supply Curve SS at the point E. If we see further, at the Price level $0P_1$ the Quantity demanded is OQ_1 and the quantity supplied is OQ_2. At this point the supply is greater than demand ($OQ_2 > OQ_1$), creating a situation of Excess Supply. This will lead the prices to fall up till the point where the demand curve intersects the Supply Curve.

Similarly, at the Price Level OP_2 the Quantity demanded is OQ_2 and the quantity supplied is OQ_1. At this point the demand is greater than the supply ($OQ_2 > OQ_1$), creating a situation of Excess Demand. This will lead to the prices to rise till the point where the Demand Curve intersect the Supply curve and the market is at equilibrium.



Examiner's Comment

- Most of the candidates did not write the definition of equilibrium. In many answer scripts the diagrams were: incorrect / not properly labelled/ direction of arrows not shown/ excess demand and excess supply not shown/ demand and supply curves wrongly drawn. Some candidates, who drew the correct diagram to show equilibrium price, did not discuss the manner in which the equilibrium was achieved.

Answering Tip

- The students should focus on the technique of how to write 'Equilibrium'. Learn with the help of a schedule and a graph, separately showing the excess demand and excess supply. Use the demand curve and supply curve to attain the equilibrium price. On the demand side price is shown because of marginal utility and on supply side price is shown because of marginal cost. The area of excess supply and excess demand should be shown in the diagram and the manner in which equilibrium is achieved.

Q.2. What is meant by floor price? Explain its impact on producers. **[ISC 2017]**

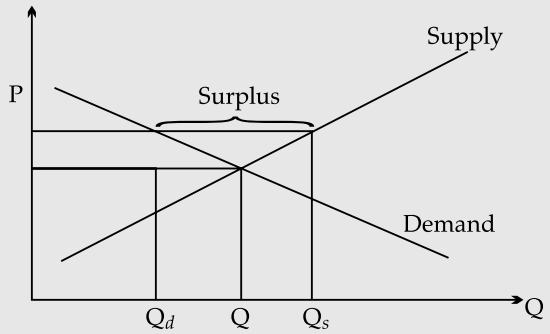
Ans. A price floor is the lowest legal price a commodity can be sold at. Price floors are used by the Government to prevent prices from being too low. The most common price floor is the minimum wage--the minimum price that can be paid for labour. Price floors are also used often in agriculture to try to protect farmers.

For a price floor to be effective, it must be set above the equilibrium price. If it's not above equilibrium, then the market won't sell below equilibrium and the price floor will be irrelevant.

- Price Floor (a minimum price or wage)
- Creates a surplus if above equilibrium

–1) Greater Q_s at the higher price

–2) Lower Q_d at the higher price



Impact on Producers: Since the price is higher than what it would be at equilibrium, the suppliers (producers) are willing to supply more than the equilibrium quantity. They will supply where their marginal cost is equal to the price floor, or where the supply curve intersects the price floor line. The decrease in demand and Increase in supply due to the new imposed higher price creates a surplus of the product.

[ISC Marking Scheme, 2017]

Examiner's Comments

- A number of candidates answered this part correctly.
- Some common errors observed in this part were incorrect definition, legal price or price fixed by government was missing, and the floor price was not shown, surplus of product as an impact of floor price was not written.

Answering Tips

- Learn the concept of floor price through a well labelled diagram.
- Focus on the impact of floor price.

Q.3. Show with the help of diagrams, the effect on equilibrium price and quantity when: **[ISC 2015]**

(i) There is fall in price of substitute goods.

(ii) There is a rise in prices of inputs.

Q.4. Explain the concept of 'Buffer stock' as a tool of price floor. **[OEB]**

Ans. Buffer stock is an important tool in the hands of government to ensure price floor/minimum support price. If in case, the market price is lower than what the government feels should be given to the farmers/ producers it would purchase the commodity at higher price from the farmers/ producers so as to maintain stock of the commodity with itself to be released in case of shortage of the commodity in future.

Q.5. Why does the government of India fix 'support price' for some crops ? Explain. **[OEB]**

Ans. When government fixes price of a product at a level higher than equilibrium price, it is called 'support price'. It is the minimum price the producers must be paid for their product. Support price is fixed for some crops to safeguard the interest of producers. Basically the main aim of fixing support price is to insulate and protect the farmers from fluctuations in their income which is caused by price variations in the free market.

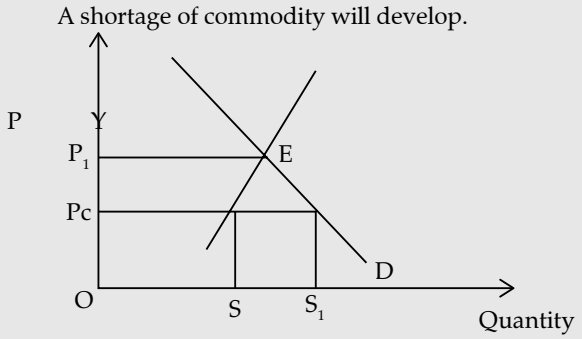
4 Marks Questions

Q.1. Explain Price Ceiling with the help of a diagram. **[ISC 2023]**

OR

Explain the concept of Maximum Price Legislation with the help of a diagram. **[ISC 2018]**

Ans. Maximum Price Legislation: It is the maximum legal price (Government price) which supplier can charge for a particular good or service. If price ceiling is set above equilibrium price it has no effect on price, there is excess supply which pulls down price. If the government fixes price below equilibrium price, it will result in excess demand. A shortage of commodity will develop.



[ISC Marking Scheme, 2018]

Examiner's Comment

- Many candidates, in the diagram, showed the maximum legislated price above the equilibrium price, which was not correct as it indicated floor price. Some candidates showed two prices - one above the equilibrium and another below equilibrium but did not indicate which of the two was the maximum legislated price.

Answering Tips

- The students need to analyse that there will be excess demand only when prices are fixed below the equilibrium price.
- The students should focus on the effects of price if set above the equilibrium level.

Q.2. What is minimum price ceiling? Explain its implications. **[OEB]**

Ans. For certain goods and services, government sets minimum price. This minimum price is called minimum price ceiling.

This price is normally set at a level higher than the equilibrium price. This leads to excess supply. Since producers are not able to sell all they want to sell, they illegally sell the goods or services below the minimum price.

Q.3. Explain 'Black marketing' as a direct consequence of price ceiling. **[OEB]**

Ans. Black marketing may be termed as a direct consequence of price ceiling, as it implies a situation whereby the commodity, under the government's control policy, is illegally sold at a higher price than the one fixed by the government. It may primarily arise due to the presence of consumer who may be willing to pay higher price for the commodity than to go without it.

Q.4. Suppose the value of demand and supply curves of a Commodity-X is given by the following two equations simultaneously: A

Qd = 200 – 10p Qs = 50 + 15p

(i) Find the equilibrium price and equilibrium quantity of commodity X.

(ii) Suppose that the price of a factor inputs used in producing the commodity has changed, resulting in the new supply curve given by the equation:

Qs = 100 + 15p

This question is for practice and its solution is available at the end of the chapter

Analyse the new equilibrium price and new equilibrium quantity as against the original equilibrium price and equilibrium quantity. [OEB]

Ans. (i) We know that the equilibrium price and quantity are achieved at :

Qd= Qs

200 – 10p = 50 + 15p

Therefore, Equilibrium Price p = ₹ 6

And, Equilibrium Quantity

q = 200 – (10) (6) =140 units 3

(ii) If the price of factor of production has changed, then under the new conditions :

Qd = Qs

200 – 10p = 100 + 15p

25p = 100

Therefore, Equilibrium Price p = ₹4

And, Equilibrium Quantity

q = 200 – (10) (4) =160 units

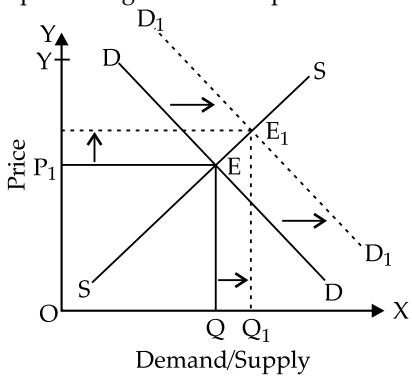
Thus, as the equilibrium price is decreasing the equilibrium quantity is increased.

Q.5. Market of a commodity is in equilibrium. Demand for the commodity 'increases'. Explain the chain of effects of this change till the market reaches equilibrium again. Also compare the prices at the old and new equilibrium. [OEB]

Ans. Suppose market is in equilibrium and demand increases. It will result in excess demand. Excess demand causes competition among buyers resulting in rise in price.

The net effect of rise in prices reduces demand and increases supply. As a result, excess demand is reduced. These changes continue till demand and supply are equal at new price.

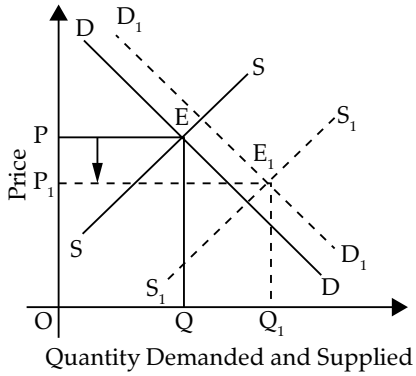
New price is higher than old price.



6 Marks Questions

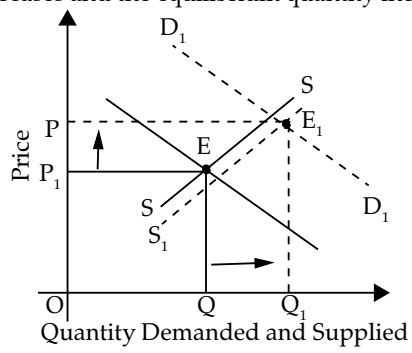
Q.1. Explain with the help of diagrams how equilibrium price changes when there is simultaneous increase of both, demand and supply. [ISC 2019]

Ans. Increase in Demand is less than Increase in Supply: When the increase in the demand is less than the increase in the supply, the equilibrium price decreases and the equilibrium quantity increases.



As can be seen in the figure, the initial equilibrium is established at point E where the initial demand curve DD intersect the initial supply curve SS. Due to the decrease in demand, due to the change in factor other than price, the demand curve shifts to the right from DD to DD_1. This leads to the change in the equilibrium level to E′. Now when the supply also increases in order meet the increase in demand, the supply curve shifts rightward from SS to S_1S_1. A new equilibrium is established at the point E_1 where the supply curve SS_1 intersect the Demand Curve DD_1. The new equilibrium price is OP_1 and the equilibrium quantity is OQ_1. As the shift in the supply is more than the increase in demand, the equilibrium price decreases to OP_1 and there is an increase in the equilibrium quantity to OQ_1.

Increase in Demand is more than increase in Supply: When the increase in the demand is less than the increase in the supply, the equilibrium price decreases and the equilibrium quantity increases.



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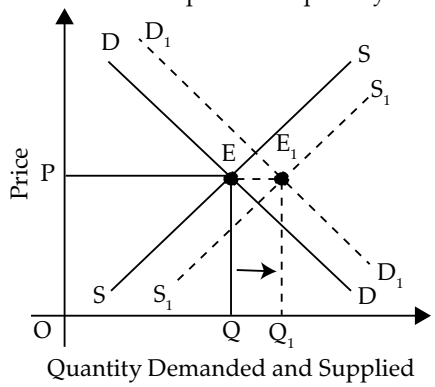

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Examiner's Comment

- Some candidates drew a single diagram instead of drawing three different diagrams with three different cases. Candidates were confused with the leftward/rightward shift of demand and supply curve. A few candidates wrote that price would change without mentioning whether it would increase or decrease. They showed changes by either keeping the demand constant or supply constant, which was incorrect.

Answering Tips

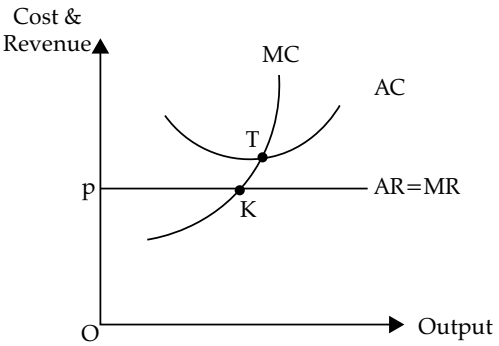
- Learn the three conditions when there is simultaneous increase of both demand and supply i.e. when price rises, when price remains same and when price falls.
- Emphasize on the fact that proper shift of demand and supply curve should be shown with the help of arrows.
- Stress that the effect of changes in price and quantity should be written.

Q.2. Explain how equilibrium price can be determined with the help of: [ISC 2016]
(i) Demand and supply schedule
(ii) Demand and supply curves

OBJECTIVE TYPE QUESTIONS

[A] Multiple Choice Questions

Q.1. Identify the equilibrium of the firm in the graph given below: [ISC Specimen Paper, 2023-24]



(A) T is the equilibrium point at which firm enjoys supernormal profit.
(B) K is the equilibrium point when firm faces losses.
(C) T is the equilibrium point causing normal profit to the firm.
(D) K indicates equilibrium which reveals zero profit to the producer

Ans. Option (B) is correct.

Q.2. A firm's demand curve shows highest elasticity in: [ISC Specimen Paper, 2023-24]
(A) Monopoly market.
(B) Perfectly Competitive market.
(C) Monopolistic competitive market.
(D) Oligopoly market.

Q.3. Which one of the following will cause a rise in the equilibrium price of rice when the demand for rice remains the same? [ISC 2023]
(A) Government subsidy to rice farmers
(B) Successful advertising campaign for rice
(C) Improved rice farming methods
(D) Poor weather conditions

Ans. Option (D) is correct.

Explanation: A decrease in supply due to poor weather condition will cause a rise in the equilibrium price of rice when the quantity demand for rice remains the same. This is because in case of poor weather conditions, supply curve will shift towards left and hence demand remaining same, equilibrium price will rise.

This question is for practice and its solution is available at the end of the chapter

Q.4. What will be the effect on equilibrium price if supply is decreased without any change in demand? [ISC Semester-I, 2022]
(A) No change.
(B) It will fall.
(C) It will rise.
(D) It will move in circular manner.

Ans. Option (C) is correct.
Explanation: If the supply of a commodity decreases, while demand remains constant, equilibrium price will increase. Chain effect of Excess Demand takes place. Due to excess demand, there will be competition among consumers, leading to rise in price.

Q.5. What will be the effect on equilibrium price when increase in demand equals increase in supply? [ISC Semester-I, 2022]
(A) Equilibrium price remains constant.
(B) Equilibrium price rises.
(C) Equilibrium price falls.
(D) Equilibrium price rises and falls respectively.

Ans. Option (A) is correct.
Explanation: When increase in supply is equal to increase in demand, the price will remain the same and the equilibrium output will increase.

Q.6. What would the price ceiling lead to, when maximum price is fixed below the equilibrium price? [ISC Semester-I, 2022]
(A) Excess demand. **(B)** Excess supply.
(C) Deficient demand. **(D)** No change at all.

Ans. Option (A) is correct.
Explanation: A maximum price below the equilibrium price results in excess demand which, in turn, invites a black market. A black market is that market situation in which goods are sold at a price more than the price fixed by the government.

Q.7. Which of the following is an example of agricultural support price? [ISC Semester-I, 2022]
(A) Price ceiling. **(B)** Floor price.
(C) Equilibrium price. **(D)** Product price.

Ans. Option (B) is correct.
Explanation: Support price or minimum price is also called Floor Price. Minimum support price is a law or regulation which holds the market price above the equilibrium price. Minimum Support Price helps to protect the interest of farmers who grow sugar cane, wheat, etc. This policy helps the farmer to sell whatever they produce and guarantee them a minimum income.

Q.8. A monopoly market has: [ISC Semester-I, 2022]
(A) No barriers to entry. **(B)** Many substitutes.
(C) Many suppliers. **(D)** One seller.

Ans. Option (D) is correct.
Explanation: A monopoly market has 1. One seller 2. No close substitutes 3. Has Barriers to entry

Q.9. Selling cost is not required in: [ISC Semester-I, 2022]
(A) Perfect competition and monopolistic competition.
(B) Oligopoly and monopolistic competition.
(C) Monopoly and oligopoly.
(D) Monopoly and perfect competition.

Q.10. Which of the following is not a characteristic of a price taker firm? [ISC Semester-I, 2022]
(A) TR = P × Q. **(B)** AR = Price.
(C) AR = MR = Price.
(D) Negatively sloped demand curve.

Ans. Option (D) is correct.
Explanation: The perfectly competitive firm is a 'price-taker' and can sell any amount of the commodity at the established price. Therefore Demand curve is infinitely elastic and given by a horizontal line. Here Price = AR = MR. And, AR curve is also the MR curve of the firm. TR will be the product of Quantity sold at the given price.

Q.11. In the context of monopolistic competition, which of the following is correct? [ISC Semester-I, 2022]
(A) Firm has full control over price.
(B) Horizontal straight line in demand curve of the firm.
(C) Freedom of entry and exit.
(D) Selling cost do not exist.

Ans. Option (C) is correct.
Explanation: Monopolistic Competition is characterised by following features:
(i) Large number of buyers and sellers: Each firm is supplying a small percentage of total market supply of the product. so no control over prices
(ii) Differentiated products: there are many close substitutes for each product and thus, a monopolistic firm faces an elastic demand curve sloping down
(iii) Free entry and exit: Firms can freely move in and out of industry as there are no barriers 4. Heavy selling cost are incurred- for promoting the demand for its product.

Q.12. If revenue equals cost, then firm earns only: [ISC Semester-I, 2022]
(A) Normal profit **(B)** Pure profit
(C) Economic profit **(D)** Supernormal profit

Ans. Option (A) is correct.
Explanation: When TR = TC or AR = AC, the firms earns only Normal Profits. Since profit equals total revenue minus total cost, normal profit means zero economic profits.

Q.13. Shut down point is a situation where the firm covers: [ISC Semester-I, 2022]
(A) Only fixed cost. **(B)** Only variable cost.
(C) Both fixed and variable cost.
(D) Neither fixed nor variable cost.

Ans. Option (B) is correct.
Explanation: At the shut-down point since AR is less than AVC the firm should stop producing. Because, if price falls below the price at the shutdown point, then the firm will be not even covering its variable cost therefore shutting down can reduce variable costs to zero. Since at a price above the shutdown point, the firm is making enough revenue to cover at least a portion of fixed costs, but at shutdown point it covers only the variable cost.

[B] Assertion-Reason Questions

Directions: In the following questions, a statement of Assertion (A) is followed by a statement of Reason (R). Mark the correct choice as:

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

This question is for practice and its solution is available at the end of the chapter

Q.1. Assertion: Price discrimination is possible only in a monopoly market.

Reason (R): AR curve of a monopoly firm exhibits the market demand for the commodity.

[ISC Specimen Paper, 2023-24]

Ans. Option (B) is correct.

Explanation: Due to the presence of no substitute there is the possibility of price discrimination. This makes the assertion to be true. AR curve does exhibit the market demand for the commodity as there is only one seller. But this is not the reason for having price discrimination.

Q.2. Assertion (A): The vegetable market is a perfect example of perfect competition market.

Reason (R): The firms have no control over the prices of the product. **[OEB]**

Q.3. Assertion (A): There is no restriction on the entry and exit of the firms in the perfect competitive market.

Reason (R): The perfect competition market is characterised by the sellers being a price taker and not a price maker. **[OEB]**

Ans. Option (B) is correct.

Explanation: Both Assertion and Reason are the characteristics of a perfect competition market.

Q.4. Assertion (A): In monopolistic competition there is a fierce competition between the firms in the market.

Reason (R): The monopolistic competition there are a large number of seller and buyers in the market. **[OEB]**

Ans. Option (B) is correct.

Explanation: Monopolistic Competition is found in the industry where there are large number of sellers selling differentiated product to a large number of buyers. There is freedom of entry and exit for the firms. In such a market, a seller has a partial control over price through product differentiation. However, full control over price is not possible owing to the fact that there are large number of close substitutes in the market, leading to a fierce competition.

Q.5. Assertion (A): In oligopoly market the firms restrict the entry of new firms.

Reason (R): Oligopoly market has few sellers which influence the decisions of the market and restrict the entry of new firms. **[OEB]**

Ans. Option (a) is correct

Explanation: There are various barriers to the entry of new firms. These barriers are almost similar to those under monopoly. Patent rights is the most important form of entry-barriers. Entry of the new firms is extremely difficult, if not impossible.

Case Based Questions

[A] Multiple Choice Questions

I. Pre COVID, an oxygen cylinder could be refilled for ₹ 300. During the second wave of the pandemic, the people bought oxygen cylinder and stored it. Hence, the price of one refill went up to ₹ 2,000. Answer the following questions:

[ISC Semester-I, 2022]

(a) The reason for the price increase was:

(A) The demand for oxygen increased and supply for oxygen remained the same.

(B) The demand for oxygen and the supply for oxygen both increased simultaneously.

(C) The demand for oxygen and the supply for oxygen decreased simultaneously.

(D) The demand for oxygen and the supply for oxygen remained unchanged.

Ans. Option (A) is correct.

Explanation: When demand of a commodity increases, while supply remains constant, equilibrium price will increase. Due to excess demand, there will be competition among consumers, leading to rise in price that's why during second wave refill price increased upto ₹ 2000.

(b) It is a case of _____ in demand curve.

(A) Leftward shift of demand curve.

(B) Upward movement along the demand curve.

(C) Downward movement along the demand curve.

(D) Rightward shift of demand curve.

Ans. Option (D) is correct.

Explanation: Keeping supply constant, if the demand increases, the demand curve shifts upwards or rightwards.

(c) People bought oxygen cylinder and stored it.

(A) The above is a case of price ceiling.

(B) The above is a case of hoarding.

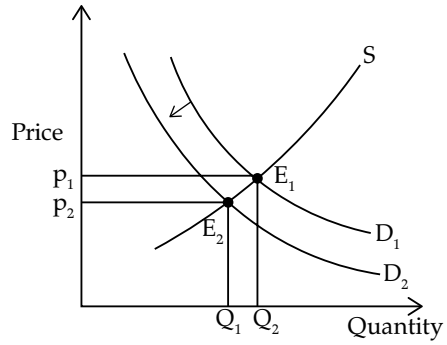
(C) It is a case of floor price.

(D) It is a case of rationing.

Ans. Option (B) is correct.

Explanation: Producers will start hoarding the goods if their price increase, to create artificial scarcity in the market.

II. As per the diagram given above:



[ISC Semester-I, 2022]

This question is for practice and its solution is available at the end of the chapter

(a) Equilibrium price has shifted from:

(A) OP_1 to OP_2 **(B)** E_1 to E_2
(C) OQ_1 to OQ_2 **(D)** OP_2 to OP_1

Ans. Option (A) is correct.

Explanation: If the demand of a commodity decreases, while supply remains constant, the equilibrium price and output will fall. The equilibrium price falls from OP_1 to OP_2.

(b) Equilibrium quantity has shifted from:

(A) OP_2 to OP_1 **(B)** E_2 to E_1
(C) OQ_1 to OQ_2 **(D)** D_1 to D_2

Ans. Option (C) is correct.

Explanation: If the demand of a commodity decreases, while supply remains constant, the equilibrium price and output will fall. The equilibrium price falls from OP_1 to OP_2 and output falls from OQ_1 to OQ_2.

(c) The new equilibrium point is:

(A) E_2 **(B)** E_1
(C) P_1 **(D)** Q_1

Ans. Option (A) is correct.

Explanation: If the demand of a commodity decreases, while supply remains constant, the equilibrium price and output will fall. The equilibrium price falls from OP_1 to OP_2 and output falls from OQ_1 to OQ_2 The new equilibrium is established at point E_2.

III. When a soft drink company A roped in a Hollywood star for its endorsement in India, the soft drink company B went in for a Bollywood star as a competitor. [ISC Semester-I, 2022]

(a) This is a feature of:

(A) Differentiated products.
(B) Non price competition.
(C) Interdependence and non-price competition among firms.
(D) Price discrimination.

Ans. Option (C) is correct.

Explanation: Under Oligopoly, since the market is dominated by a few firms, the price and output decisions of one firm affects the profitability of the remaining firms in the market. Mutual interdependence is an incentive to develop alternatives to price competition in pursuit of economic profit. Each firm carefully considers and watches how its actions will affect its rivals and how its rivals are likely to react. Firms try to avoid price competition for the fear of price war.

(b) Name the kind of market:

(A) Monopoly.
(B) Perfect competition.
(C) Oligopoly.
(D) Monopolistic Competition.

Ans. Option (C) is correct.

Explanation: Oligopoly is a market situation in which an industry has only a few firms (or few large firms producing most of its output) mutually dependent for taking decisions about price and output.

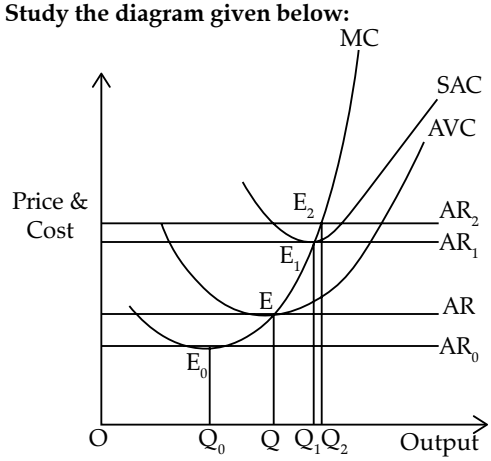
(c) The number of sellers in this kind of market are:

(A) Infinite. **(B)** Few.
(C) Only one.
(D) Large number but not infinite.

Ans. Option (B) is correct.

Explanation: There are many sellers in Perfect competition and Monopolistic competition market. There is only one seller in the case of Monopoly market. Oligopoly market is characterised by the presence of only few sellers in the market. Oligopoly is a form of market in which there are few big sellers of a commodity and a large number of buyers. Each seller has a significant share of the market.

IV. Study the diagram given below:



[ISC Semester-I, 2022]

(a) Point E is:

(A) Shut down point **(B)** Break – even point
(C) Market equilibrium point
(D) Point of minimum return

Ans. Option (A) is correct.

Explanation: Point 'E' in the diagram is shutdown point because here AR (Price) = AVC = MC. In other words, the point where the marginal cost curve crosses the average variable cost curve it gives the shutdown point. When the market price (AR) that a perfectly competitive firm faces is below average variable cost (AVC) at the profit-maximising quantity of output, then the firm should shut down operation.

(b) Point E_1 is:

(A) Shut down point
(B) Break – even point
(C) Market equilibrium point
(D) Point of maximum return

Ans. Option (B) is correct.

Explanation: Point E_1 shows Breakeven Point. When the market price (AR1) is equal to average cost (SAC) at the profit-maximizing level of output, at this point the firm is making zero profits and it's the breakeven point. Similarly the point where the MC curve crosses the average cost (AC) curve, at the minimum of the average cost curve, it determines the break-even point.

(c) At E_0:

(A) Firm may stop producing
(B) Firm must stop producing
(C) Firm may continue production
(D) Firm must continue production

Ans. Option (B) is correct.

Explanation: At E_0 point, since AR < AVC the firm should stop producing. As we know, if price falls below the price at the shutdown point, then the

firm will shut down immediately, since it is not even covering its variable costs.

(d) Point E_2 indicates:

(A) Normal profit **(B)** Loss
(C) Supernormal profit
(D) Neither loss nor profit

Ans. Option (C) is correct.

Explanation: Point E_2 indicates Super normal profits as at this point AR > AC. Supernormal profit is made where average revenue exceeds average cost. In a perfectly competitive market, firms are price takers facing infinite demand at the market price which means that average revenue is not only equal to marginal cost, but is also constant and equal to the market price. Costs on the other hand are kept as low as possible to avoid being priced out of the market. Output is determined by the profit maximising condition i.e. where marginal revenue is equal to marginal cost. This could potentially lead them to a situation where average cost exceeds average revenue but this will only be the case in the short run.

[B] Subjective Based Questions

I. More specifically, a price ceiling (in other words, a maximum price) is put into effect when the government believes the price is too high and sets a maximum price that producers can charge; this price must lie below the equilibrium price in order for the price ceiling to have an effect.

The price ceiling is usually instituted via law and is typically applied to necessary goods like food, rent, and energy sources in order to ensure that everyone has access to them.

Benefits and Downsides:

Price ceilings are beneficial to society, and are often necessary, in that they make sure that essential goods are financially accessible to the average person, at least in the short run. By lowering costs, price ceilings also have the beneficial effect of helping to stimulate demand, which can contribute to the health of an economy.

However, there can also be downsides to price ceilings. While they stimulate demand, price ceilings can also cause shortages. Where the ceiling is set, there is more demand than at the equilibrium price. This means that the amount of the good or service supplied is less than the quantity demanded.

For example, in agriculture, medicine, and education, many governments set maximum prices to make the needed goods or services more affordable. Producers may respond to such an economic situation by rationing supplies, decreasing production levels or lowering the quality of production, making the consumer pay extra for otherwise free elements of the good (features, options, etc.), and more. **[OEB]**

- Price Ceiling – Intelligent economists.

(a) When does the government put price ceiling? Explain.

Ans. The government puts price ceiling where it believes that prices are too high. This is done in order to avoid the exploitation of people.

(b) Too which items is the price ceiling applied to?

Ans. Price ceiling is typically applied to necessary goods like food, rent, and energy sources in order to ensure that everyone has access to them.

(c) How does price ceiling help stimulate demand?

Ans. When cost or price is lowered, price ceilings have the beneficial effect of helping to stimulate demand which leads to healthy economy.

(d) How do producers react to the situation of price ceiling?

Ans. The producers respond to the situation of price ceiling by rationing supplies, reducing the production level and also by lowering the quality of production.

(e) Does price ceiling exists above or below the equilibrium price? And does it create surplus or shortage?

Ans. Price ceiling exists below the equilibrium price. Hence, it leads to shortage because demand is greater than supply. When demand is greater than supply, then it leads to excess demand. A price ceiling does not cause price to rise further.

VI. A price floor is the lowest legal price that can be paid in a market for goods and services, labour, or financial capital. Perhaps the best-known example of a price floor is the minimum wage, which is based on the normative view that someone working full time ought to be able to afford a basic standard of living. The federal minimum wage at the end of 2014 was $7.25 per hour, which yields an income for a single person slightly higher than the poverty line. As the cost of living rises over time, Congress periodically raises the federal minimum wage.

Price floors are sometimes called price supports because they support a price by preventing it from falling below a certain level. Around the world, many countries have passed laws to create agricultural price supports. Farm prices, and thus farm incomes, fluctuate—sometimes widely. So even if, on average, farm incomes are adequate, some years they can be quite low. The purpose of price supports is to prevent these swings.

The most common way price supports work is that the government enters the market and buys up the product, adding to demand to keep prices higher than they otherwise would be.

—Source: Khan Academy

(a) Which is lowest legal price?

Ans. Price floor is the lowest legal price that can be paid in a market for goods and services, labour, or financial capital.

(b) Explain the best known example of price floor.

Ans. Minimum wage is the best-known example of a price floor. The federal minimum wage at the end of 2014 was $7.25 per hour, which yields an income for a single person slightly higher than the poverty line. As the cost of living rises over time, Congress periodically raises the federal minimum wage.

(c) "Price floor is also called price support." Explain.

Ans. Price floor is also known as price support because it prevents price from falling below the certain level. Many agricultural rice supports have been created and laws have been passed on them.

(d) Why does price support prevent price swings?

Ans. Farm prices, and thus farm incomes, fluctuate—sometimes widely. So even if, on average, farm

incomes are adequate, some years they can be quite low. The purpose of price supports is to prevent these swings

(e) Does price floor exists above or below the equilibrium price? And does it create surplus or shortage?

Ans. Price floor exists above the equilibrium price. And it leads to surplus because above the equilibrium price, supply is more than demand. And hence excess supply is created or results in surplus. A price floor does not cause price to fall further.

Solutions for Practice Questions (Topic-1)

1 Mark Question

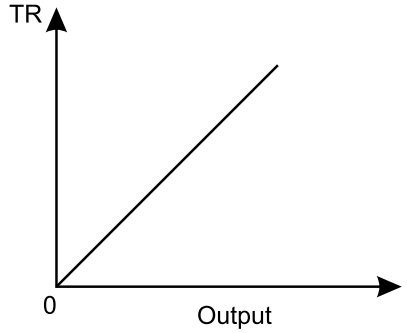
2. Monopoly refers to that form of market organisation wherein there is a single firm or producer producing a commodity for which there are no close substitutes. Example: Indian railways, state electricity board.

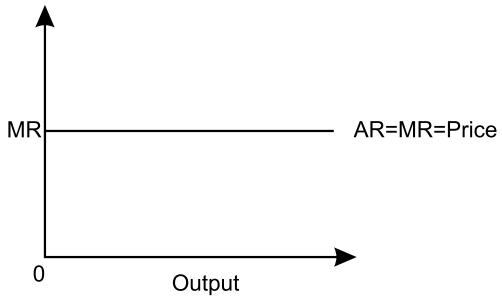
[ISC Marking Scheme, 2017]

3 Marks Questions

3. In Perfect Competition Market the firm does not have to change its price to sell more.

The TR curve and MR Curve are:





4 Marks Questions

2. The difference between oligopoly market and monopolistic competition market are:

S. No.	Basis	Oligopoly Market	Monopolistic Competition
(1)	Number of buyers and sellers	There are a large number of buyers and few sellers.	There are a large number of buyers and sellers.
(2)	Demand Curve	The firm faces an in determinant demand curve.	The firm faces an elastic demand curve.
(3)	Free entry and exit.	There is restricted entry.	There is free entry and exit.
(4)	Competition	Cutthroat competition.	Less competition due to monopoly in brand name.

6 Marks Questions

3. The features of perfect competition market are:

(i) Very Large number of buyers and sellers: In perfect competition, there must be large number of buyers and sellers. Each buyer buys a small quantity of the total amount. Each seller is so large that no single buyer or seller can influence the price and affect the market. Buyers and sellers are price takers in the purely competitive market. Each seller sells its products at the price determined by the market. Similarly, each buyer buys the commodity at the price determined by the market.

(ii) Homogenous Product: In this case, all sellers produce homogeneous i.e. perfectly identical products. All products are perfectly same in terms of size, shape, taste, colour, ingredients, quality, trademarks etc. This ensures the existence of single price in the market.

(iii) Free entry and exit: Under perfect competition, there will be no restriction on the entry and exit of both buyers and sellers. If the existing sellers start making abnormal profits, new sellers should be able to enter the market freely. This will bring down the abnormal profits to the normal level. Similarly, when losses will occur existing sellers may leave the market. However, such free entry or free exit is possible only in the long run, but not in the short run.

(iv) Perfect knowledge: Perfect competition implies perfect knowledge on the part of buyers and sellers regarding the market conditions. As a result, no buyer will be prepared to pay a price higher than the prevailing price. Sellers will not charge a price higher or lower than the prevailing price. In this market, advertisement has no scope.

(v) **Price Taker:** There are large number of sellers in perfectly competitive market, so that an individual firm has a negligible share in total supply. As such no individual seller can influence the market price on its own. The seller has no option but to accept the market determined price. It makes the seller a 'Price Taker'.

(vi) **Absence of transport cost:** Under perfect competition transport, cost does not exist. Since commodities have, the same price it logically follows that there will be no transport cost. In the event of the presence of cost of transport, there will be no single price in the market. Transport cost occurs when there is no perfect knowledge of the market conditions on the part of buyers and sellers.

(vii) **Absence of Selling Costs:** There is no need for a firm working under perfect competition to undertake selling costs, since by assumption, the product produced by all firms in the perfectly competitive industry is homogeneous, and an individual firm can sell as much quantity of the product as it likes at the given price.

(viii) **Perfect Mobility of the Factors of Production:** There should be perfect mobility of factors between industries. Factors should be free to move to those places where they can fetch the highest price.

(ix) **Horizontal average and marginal revenue curves:** In such a market whether the firm sells more or less, it does not affect the market price. It makes the AR curve parallel to the X-axis so at all levels of quantity produced, market price remains unchanged. Since AR is unchanged, AR must be equal to MR. The MR curve thus coincides with the AR curve. **[Any Four]**

Examiner's Comment

- Many candidates did not answer this part correctly. A few candidates only wrote the captions.

Answering Tip

- Be careful with the headings and the explanations.

Solutions for Practice Questions (Topic-2)

1 Mark Questions

4. It can be defined as the point at which firm is not even able to cover average variable cost of production or AVC < AR.

2 Marks Questions

3. **Break-even point:** It is the point at which AC = AR. This means, it is the point at which the firm is neither making any profits nor incurring any losses.

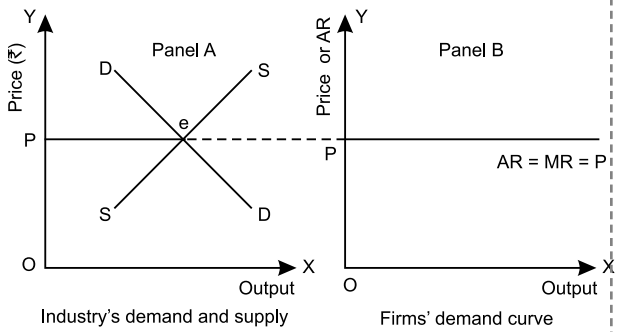
   Shut down point: It is the point at which the firm is not even able to recover per unit variable cost on sale of a single unit of the commodity or, AVC > AR.

6 Marks Questions

3. Under perfect competition, the individual firms sell homogenous products. Since the products are homogenous, the consumer can purchase goods from any seller. Also, there are large number of buyers and sellers in this type of market. The number of sellers is so large that a single seller cannot determine individual supply curve (due to mobility of the consumer to other buyer) nor can the individual buyer (with a small share of total market output) influence price of the commodity. On the other hand, industry as a whole can determine demand and supply of the commodity in the market.

   Hence, the industry determines equilibrium prices by market forces of demand and supply. This equilibrium price is taken up by individual firms as the constant price on which each unit of the commodity is sold to the consumers.

   This has been illustrated in this diagram also.



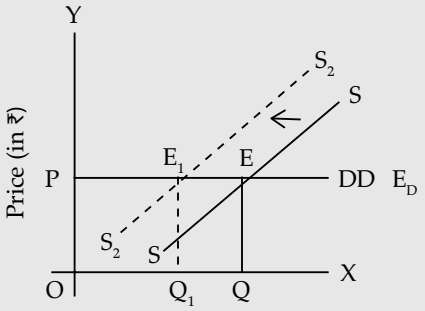
Solutions for Practice Questions (Topic-3)

1 Mark Questions

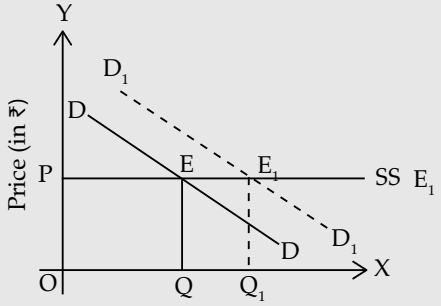
4. When supply curve is perfectly inelastic, then equilibrium price rises but equilibrium quantity remains same.

2 Marks Questions

2. **(i)** Decrease in supply when demand is perfectly elastic:



(ii) Increase in demand when supply is perfectly elastic:



Quantity Demanded and Supplied (in units)

[ISC Marking Scheme, 2018]

Examiner's Comments

- While explaining the meaning of full employment, several candidates did not mention the term prevailing wage rate. Some candidates wrote any wage rate or wage rate, both being incorrect.
- While explaining the meaning of involuntary unemployment, many candidates either did not mention the term prevailing wage rate or wrote that people failed to get the job at satisfactory wage rate.

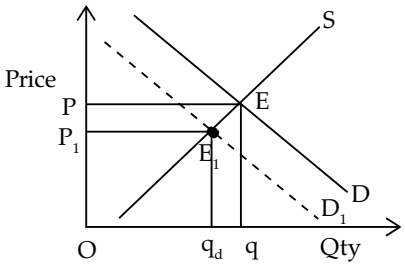
Answering Tip

- The students need to understand that for an increase or decrease in supply, the indication on the curve is a must which is either by a right arrow or by a left arrow.

3 Marks Questions

3. **(i)** There is fall in price of substitute goods.

Fall in price of substitute will shift demand curve to the left.

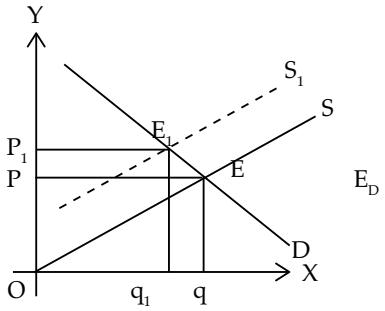


Equilibrium shifts E to E1.

e.g., price from P to P1 and Eq quantity q and q1

(ii) Rise in prices of inputs.

Rise in price of inputs will shift the supply curve to the left.



The equilibrium shifts from E to E1

e.g. Quantity from q and q1

e.g. Price from P to P1

OR Self-explanatory diagram

Examiner's Comments

- Some candidates depicted shift in both demand and supply curves whereas only demand curve should have shifted to the left.
- In several cases, the demand curve was shifted which was incorrect. In some cases, both demand and supply curves were shifted.

Answering Tip

- The students need to practice to draw properly labelled diagrams for equilibrium positions.

6 Marks Questions

2. Meaning: Equilibrium price is that price at which its two determinants e.g. demand and supply are in balance i.e. demand is equal to supply.

Schedule:

Price (₹ per kg)	Demand (in kg)	Supply (in kg)	Remark
10	30	18	Excess Demand
11	28	20	
12	26	22	
13	24	24	Equilibrium
14	22	26	Excess Supply
15	20	28	
16	18	30	

Diagram refer to Q1 of 3 Mark Question Topic 3

[ISC Marking Scheme, 2016]

Examiner's Comments

- Most of the candidates are not able to show equilibrium with the help of schedules. They did not show excess demand and excess supply in both cases.
- In a number of cases, the demand and supply curves were wrongly labelled. Excess supply and excess demand were not shown.

Answering Tips

- Students should understand the meaning of equilibrium as "stability" where quantity demanded= quantity supplied. In this context, excess demand and excess supply "instability" needs to be explained.
- Equilibrium means "stability". How to achieve equilibrium (stability) from instability (excess demand and excess supply) needs to be stressed upon.

Solutions for Objective Type Questions

[A] Multiple Choice Questions

2. **Option (B) is correct.**
9. **Option (D) is correct.**

Explanation: There is no need to incur selling cost for a firm working under perfect competition or having Monopoly in the market since by assumption, the product produced by all firms is homogeneous in the perfectly competitive industry and by a Monopoly firm, and individual firm can sell as much quantity of the product as it likes at the given price.

[B] Assertion Reason Questions

2. **Option (A) is correct.**

Explanation: As the firms in the vegetable market are price takers that is they do not control the prices of the product they are a perfect example of perfectly competitive market.

REFLECTION

(a) Can you differentiate between all the types of markets?

(b) Can you, with the help of a diagram, explain the shut down point and break – even points for the perfect competition firm?

(c) Can you establish market equilibrium when either demand changes or supply changes or both at various degrees of elasticity?

(d) Do you understand how does price ceiling and price floor affect the market mechanism?